

Project #7 - Getting to know Python

Learning objectives:

- Design and implement a complete program in an interpreted language.
- Use Classes, Iterators in Python.
- Perform file and console I/O in Python.

Overview:

The objective of this assignment is to get to know Python classes, Iterators, and I/O. You will write a “Student Records” program. Reading from a provided text file and writing to the standard output and to a new text file.

Specifications and suggestions:

When I run your program, it should produce a listing of the student records as follows:

```
Student Records
=====
UID:
First Name:
Last Name:
Level:
CLASSES:
CS3100
CEG2170
.....
.....
UID:
First Name:
Last Name:
Level:
CLASSES:
CEG3310
.....
.....
```

- When executed, your program should look for the file `studentRecordsIn.txt` (provided) in the same directory as your main program script. It should load the list of records from this file. If this file is not present, your program should exit gracefully (with a message).
- Each line is a full student record with the last words are for the classes:
 - Student record order: UID, FIRST, LAST, LEVEL, CLASSES
- After reading records, your program should output the data to **console** formatted as described above, and then print the output to **text file** `studentRecordsOut.txt` in the same format, and save it in the same directory.
- Your program must have at least one defined **class** "Student" to hold a student record, a method **add_class**, a method **__str__**, and an **iterator**, **__iter__** and **__next__** methods. You may define more classes and methods as you need.
- When a new Student object record is created, with (uid, first, last, level), and classes, calling the print (new_student) will print new record to **console** and append to the **output txt file**.

For example: If I add the following code to the end of your program it will output to **console** the two new records after the initial five records, and append the **output text file** with the two new records.

```
st1 = Student(uid, first, last, level);
```

```
st1.add_class(class1)
```

```
st1.add_class(class2)
```

```
st1.add_class(class3)
```

```
st2 = Student(uid, first, last, level);
```

```
st2.add_class(class1)
```

```
st2.add_class(class2)
```

```
print(st1)
```

```
print(st2)
```

What to Turn In:

Submit, well-commented file with your Python code via the Pilot dropbox. The python filename should be **project_7.py**. Include your name in the header comments.

Grading:

This project is worth 50 points. Your program will be graded as follows:

1. **Minimum requirements (37 pts)** – your program should run correctly and read from the records file and output the data to console and to a text file formatted as described.
2. **Adding new student record: (8 pts)** – when adding new student record, your program should print new record to **console** and append the **output text file**. (see above example)
3. **Style and comments (5 pts)** –Your program should be organized, **well-commented** with class and method headers.